1. Identity

Document	STAR-PKG-STAR-MANEUVER-EN · v2.2 · 2026-11-02
Package	star-maneuver
Version	0.2.2
Capability	star.orbit.maneuver
Maturity	Beta (VERIFIED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_maneuver

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_maneuver-0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_maneuver-0.2.2-py3-none-any.whl	ec5599d2262b3f7e0bf9b087e0b73fe513d7722c3ed37ec9a25842eff83c8e30
star_maneuver.py	c1908a47917dfdba2b61883b2d442ec547e51c6bc4986e94bb71d4ef7d8dfc83
star_maneuver-0.2.2-source.zip	b1599d00986103ca9ea5ef28d872bc7c6cdc5567e8830fc0b60493684ffe0504

5. What it does, and its limits

Impulsive manoeuvre arithmetic between circular coplanar orbits and plane changes. Standard library only.

Finite burns, gravity losses, non-coplanar Hohmann with optimal plane-change split, any flight qualification.

- `vis_viva(r, a, mu)`; `hohmann(r1, r2, mu)` and `bielliptic(r1, r2, rb, mu)` return per-burn and total delta-v (km/s) and time of flight (s). Units: km, s, km/s, km^3/s^2 (default mu = 398600.4418, IERS Conventions 2010).
- `plane_change(v, di)` = 2 v sin(di/2); `combined_dv(v1, v2, di)` by the law of cosines (radians).
- Invalid input fails explicitly with `ValueError` (non-positive radii, rb < max(r1, r2), negative speed).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_maneuver_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-maneuver}: Closed-form vis-viva and Hohmann orbital transfer engine},
```

```
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_maneuver/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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